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TAYTAY, RIZAL

**DESIGN AND DEVELOPMENT OF A QR CODE BASED SMART INVENTORY AND
ASSET TRACKING SYSTEM FOR COMPUTER LABORATORY EQUIPMENT IN
CEIT DEPARTMENT**

BSCS/BSIT - 4

A Thesis/Capstone Project Presented to
The Faculty of the College of Engineering and Information Technology
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In Fulfillment of the Requirements
For the Degree Bachelor of Science in Information Technology

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CHAPTER 1

Introduction

Background of the Study

The effective management of information technology (IT) equipment is an important part of maintaining an organized, secure, and productive educational environment. Computer laboratories contain numerous valuable assets, including computer units, monitors, keyboards, mice, headsets, networking devices, and other peripherals that are regularly used by students, faculty members, and laboratory personnel. As the number of equipment and users increases, maintaining accurate information regarding the location, condition, availability, borrowing history, and maintenance status of these assets becomes increasingly important. One technology that can support this process is the Quick Response (QR) code, a two-dimensional code that can be scanned using a mobile device or scanner to quickly retrieve stored information. In asset management, QR codes can be assigned to individual equipment to provide easier identification, monitoring, and updating of asset records.

Globally, educational institutions and organizations face challenges in managing large quantities of IT and laboratory equipment, particularly when traditional manual methods such as paper records and spreadsheets are used. Manual inventory processes can result in human errors, outdated records, misplaced equipment, and delays in updating asset information. These challenges are especially relevant in environments where equipment is frequently transferred, borrowed, repaired, replaced, or used by different individuals. Recent developments in digital laboratory management have demonstrated the potential of QR codes, RFID, mobile applications, and centralized databases to improve laboratory asset management and information retrieval. Digital



asset-tracking systems can provide more timely and reliable records by allowing personnel to identify equipment, monitor its status, and access information through a centralized platform. This shift toward smart asset management reflects the growing need for educational institutions to adopt technologies that can improve operational efficiency, accountability, and resource utilization.

The same concern can be observed in the Philippine educational sector (Queenie Tomaro 2018), where schools and higher education institutions increasingly depend on computers, networking equipment, and other ICT resources for teaching, learning, administration, and research. The importance of maintaining accurate ICT inventory records is recognized in Philippine educational policies. For example, the Department of Education emphasizes the proper management and monitoring of ICT resources, including maintaining updated records of ICT hardware and other resources. These policies demonstrate that accurate inventory, accountability, and proper utilization of technology resources are essential components of ICT management in educational institutions. As schools continue to acquire and use more technology-based resources, there is a growing need for efficient systems that can support the monitoring and management of these assets.

In the College of Engineering and Information Technology (CEIT) Department, computer laboratories are essential facilities used by students for academic activities, practical exercises, programming, networking, and other technology-related learning. The department has several laboratories, including the Mikrotik Laboratory, Mac Laboratory, and Computer Laboratory, which contain various types of equipment such as computer units, monitors, keyboards, mice, headsets, networking devices, and other laboratory tools. These resources are continuously used by students and personnel, making proper inventory and maintenance important for ensuring that the laboratories remain functional and ready for academic use.



Currently, the CEIT Department primarily uses a manual inventory process through spreadsheets or Microsoft Excel to monitor its laboratory equipment. While spreadsheets can provide a basic method of recording inventory information, they have limitations when equipment is frequently borrowed, transferred, repaired, replaced, or used by different students. Information may not always be updated immediately, resulting in records that may not accurately reflect the current status, location, or condition of an asset. Personnel may also need to spend additional time searching through multiple records to determine whether a particular piece of equipment is available, borrowed, damaged, missing, or undergoing maintenance.

Another concern involves the proper use and accountability of laboratory computer equipment. Students may borrow equipment such as keyboards, headsets, or other peripherals from the department, while laboratory computers may also be subject to unauthorized changes, including the installation of games or other software. Without an efficient tracking and monitoring mechanism, it can be difficult to determine when equipment was used, who was responsible for it, whether an asset was returned in its proper condition, or whether a computer requires inspection or maintenance. These circumstances may contribute to equipment misplacement, delayed maintenance, inaccurate inventory records, and increased workload for laboratory personnel.

Although existing studies and systems have demonstrated the usefulness of QR codes and digital asset-management technologies, there remains a need for a system specifically designed to address the operational requirements of an academic computer laboratory. Some existing solutions focus mainly on inventory recording, while others are designed for larger organizations with different asset-management requirements. There is a gap in providing a localized and integrated system that combines QR-based equipment identification with inventory monitoring, equipment condition tracking, borrowing and return records, asset location, and maintenance information



specifically for the CEIT Department. Addressing this gap can provide a more practical and suitable solution for the department's existing inventory-management challenges.

In response to these concerns, this study proposes the Design and Development of a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department. The proposed system will utilize QR codes to provide unique identification for laboratory equipment and a centralized digital platform to manage relevant asset information. Through the system, authorized personnel can more efficiently access and update information regarding equipment, including its status, condition, location, borrowing records, and maintenance activities. The system is intended to reduce dependence on manual spreadsheet-based monitoring while improving the accuracy, accessibility, efficiency, and accountability of laboratory asset management.

Therefore, the general objective of this study is to design and develop a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department that will improve the management, monitoring, and accountability of laboratory assets. The study aims to provide the department with a more reliable and organized approach to inventory management, enabling personnel to monitor equipment more efficiently, support timely maintenance, track asset movement, and maintain accurate records of computer laboratory resources.

Statement of the problem



This study focuses on the Design and Development of a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department. It aims to address the limitations of the current manual inventory process and improve the monitoring, tracking, maintenance, and accountability of computer laboratory equipment. Specifically, it seeks to:

- Identify and analyze the existing challenges encountered by the CEIT Department in managing computer laboratory equipment using manual inventory methods, particularly in terms of inaccurate or outdated records, equipment misplacement, borrowing and return monitoring, equipment condition, and maintenance tracking.
- Design and develop a QR code-based smart inventory and asset tracking system that can provide efficient equipment identification, centralized inventory records, asset movement monitoring, and timely updates of equipment information.
- Determine and integrate essential system features that can improve the management and usability of laboratory equipment, including QR code scanning, borrowing requests and return management, equipment status and condition monitoring, maintenance records, user accountability, and inventory reporting.
- Evaluate the effectiveness and usability of the developed system in improving inventory management, reducing errors in equipment records, increasing asset



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accountability, and making the monitoring and maintenance of computer laboratory equipment more efficient.

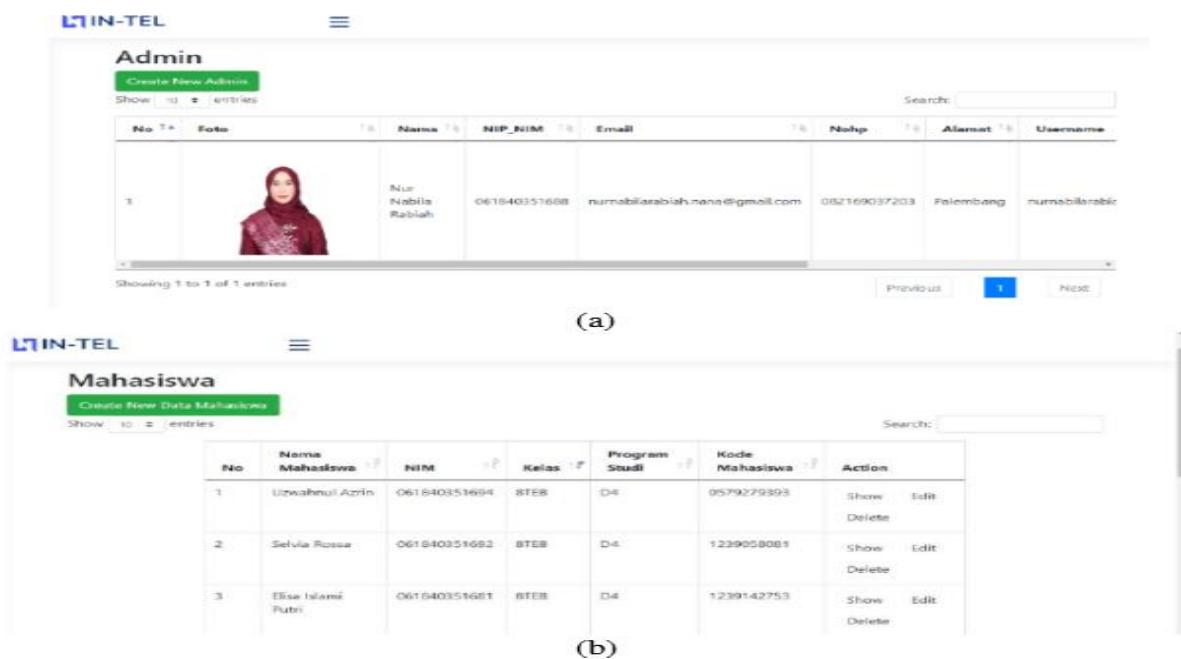
Review of Related Literature



Web-Based Laboratory Inventory Application Using QR Code and RFID

Rabiah, Lindawati, and Sarjana (2022) developed a web-based laboratory inventory application using QR Code and Radio Frequency Identification (RFID) for the Telecommunication Engineering laboratories and workshops at the State Polytechnic of Sriwijaya. The study was developed in response to the laboratory's manual equipment borrowing process, where students had to fill out loan forms and technicians had to manage inventory records manually. The researchers proposed a web-based application to provide stock information and facilitate equipment borrowing and inventory management.

Figure 1. Screenshot of the Existing Web-Based Laboratory Inventory Application



The system provides different functions for administrators, students, and laboratory technicians. The application includes tools and component data, QR-code identification, tool lending, component retrieval, and transaction recording. The tool data menu displays information about laboratory equipment and includes a QR code for identification. In the borrowing process, students provide their identification and the technician scans the QR code of the required equipment. The system records the borrowing transaction and can generate reports in PDF format.

Strengths of the Existing System

The system produce different advantages for laboratory inventory management. First a QR code that allows equipment to be identified quickly through scanning rather than just manually finding inventory records.



Second is the application focus in equipment information in a web-based system. Third, the system supports equipment borrowing and also automatically record transactions. The study mention that QR-code scanning successfully displayed the match equipment information in the web application.

Weakness of the Existing System

While the system offers QR-code-driven inventory and borrowing capabilities, its primary emphasis is on overseeing laboratory equipment and tool borrowing within a Telecommunication Engineering lab. The system additionally utilizes RFID to identify students when they borrow items. Consequently, its features are mainly tailored to the particular laboratory processes of the researchers' organization. According to the characteristics outlined in the research, there is still a chance to create an inventory system tailored to the oversight of computer hardware, equipment details, placement, and condition within an academic department.

Research Gap

The research shows that QR-code technology enhances the identification and tracking of lab equipment. Nonetheless, the current system was crafted explicitly for Telecommunications Engineering labs and focuses heavily on equipment lending and student identification via RFID. The suggested system stands out by emphasizing the management of computer hardware and equipment inventory in an academic department. It is designed to offer a centralized database of equipment details, encompassing computer specifications, hardware information, location, and status, while maintaining QR-code scanning for rapid equipment identification and generating QR-codes for newly added items.

Cloud-Based Inventory System with QR Tagging for Technology Workshops

Sermonia and Pabelona (2025) created a cloud-driven inventory management system featuring Quick Response (QR) tagging for tech workshops. The system was created to overcome the drawbacks of conventional inventory methods, wherein tools and equipment were counted and documented manually with inventory forms. The study indicates that manual inventory procedures demand significant time and effort, making it challenging to retrieve information. The suggested system focused on enhancing inventory management by utilizing managed equipment and user information, generating QR codes, scanning QR codes in real-time, reporting inventory, and synchronizing with a mobile application.

Figure 2. Screenshot of the Cloud-Based Inventory System with QR Tagging



INVENTORY SYSTEM Sign out

NAVIGATION
Dashboard
Users
Category
Item Categories
Brand
Locations
Equipment
REPORTS
Inventory Report
User Logs
Equipment Report

Users Welcome: Super Admin (Super Admin) April 23, 2022

Show 10 entries Search

☐	#	Name	User No.	User Level	Position	Status	Action
☐	1	Super Admin	21	Admin		Active	Edit User
☐	2	staff 1	33	Property Personnel		Active	Edit User
☐	3	Teacher2	343	Admin	testa	Active	Edit User
☐	4	Mike Semsonia		Mobile User	Student	Active	Edit User
☐	5	Richard Garcia	19024	Mobile User	Teacher	Active	Edit User
☐	6	Ryan Sta. Maria		Laboratory Staff	Lab staff	Active	Edit User
☐	7	q semsonia	HUK123	Mobile User	Teacher	Active	Edit User
☐	8	Mark Narlos	User21	Admin	Head	Active	Edit User
☐	9	Personel		Property Personnel		Active	Edit User
☐	10	John Doe	123154	Laboratory Staff	Lab Staff	Active	Edit User

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The system offers a web-based platform where administrators and people who are allowed to manage things can take care of equipment information. The Equipment Page gives users the ability to add and change details about the equipment. This includes the name of the equipment, the brand, where it is located how much it cost and when it was bought. The page also has tools for changing the status of the equipment looking at the history of the equipment and making QR codes to put on the equipment for tagging. The system can also create inventory reports.

The system also has an application that lets users see the list of equipment and scan the QR codes that are on the equipment. When a QR code is scanned the app shows all the details about the equipment. People can also note when they use the equipment and send reports about it. The mobile application works without internet after the time someone logs in. It automatically sends the data back, to the system when the internet is available again.

Strengths of the Existing System



The system offers strong benefits, for managing inventory. First it moves away from manual methods by using a cloud-based system that keeps everything in one central place. Second the use of QR codes makes it faster and easier to identify equipment. Third users can monitor equipment and inventory in time and get reports whenever needed. The system also tracks the history and current status of each piece of equipment. This helps people see if an item is working well being repaired or not functioning properly. Another key feature is the app. It works without internet. Saves data locally. Once the connection is back all information syncs automatically.

Weakness of the Existing System

The system offers inventory tracking and QR-tagging tools but it is built specifically for hard-skills technology workshops. These include Automotive, Machine Shop, Refrigeration and Air Conditioning, Electronics and Electricity. The mobile app lets users view equipment scan QR codes. Send reports. However managing inventory must be done through the web application by authorized users.

This study creates a base for using QR codes, in equipment management.. Its features and the types of equipment it supports were made for technology workshops. They don't fully match the needs of a computer-related department.

Research Gap

The study shows that cloud-based inventory management with QR tagging can improve the monitoring and tracking of equipment. Equipment registration, QR-code generation QR-code scanning, location management, equipment status monitoring, equipment history and report generation are functions provided by the system. These features are closely related to the objectives of the proposed system.

However the existing system was developed for technology workshops and covers equipment categories such as Automotive, Machine Shop, Refrigeration and Air Conditioning, Electronics and Electricity. The proposed system will instead focus on the inventory of computer hardware and equipment within a department. The system can be tailored to record computer specifications, hardware details, equipment location, equipment status and other information relevant to the department.

Therefore the study of Sermonia and Pabelona (2025) serves as a reference, for the proposed system because the study demonstrates the feasibility of integrating QR tagging, cloud-based inventory management, equipment monitoring and mobile QR-code scanning. The proposed system builds upon these concepts by adapting them to the inventory requirements of computer hardware and equipment in an academic department.

The Asset Management and Tracking System for TVET Institutions Based on Ubiquitous Computing

Sulistyo, Achmad and Purnama (2022) created an asset management and tracking system, for Technical and Vocational Education and Training institutions. The research was carried out at Politeknik Negeri Balikpapan. It aimed to solve the problems that come with asset management methods. These methods often rely on paper records and spreadsheets. The study found that managing assets manually can lead to challenges. It becomes hard to keep track of asset history, where assets are located their specifications, how they are used and who is using them. These problems grow worse when institutions handle a number of items. To fix these issues the researchers built a system that can be used on the web and mobile phones. They used QR codes to identify each asset. They also used geolocation to track the position of assets that can be moved.

Figure 3. Screenshot of the Asset Management and Tracking System

The screenshot displays a web application for managing equipment data. The main heading is "Halaman MASTER DATA ALAT Jurusan Teknik Sipil". Below this, there is a search bar and a table listing equipment items. The table has the following columns: No, Kode Item, Nama Alat, Lokasi, Tahun Pengadaan, Kondisi, Manual, Gambar, and Action. The Action column contains "Edit" and "Delete" buttons. The interface is in Indonesian and includes a sidebar with navigation options like "Home", "Pembuatan", "Daftar", "Kategori", "Detail", "Laporan", "Pengaturan", "Profil", "Pengaturan", "Pengaturan", and "Logout".

No	Kode Item	Nama Alat	Lokasi	Tahun Pengadaan	Kondisi	Manual	Gambar	Action
1	MEASUR-01	Measuring Tape 50 m	LAB Survei	2012	Baik			Edit Delete
2	MEASUR-02	Measuring Tape 50 m	LAB Survei	2012	Baik			Edit Delete
3	MEASUR-03	Measuring Tape 50 m	LAB Survei	2012	Baik			Edit Delete
4	MEASUR-04	Measuring Tape 50 m	LAB Survei	2012	Baik			Edit Delete
5	TRAC-05	Transit	LAB Survei	2012	Baik			Edit Delete

I find the system very useful because it offers a web-based asset management application and an Android application that both talk to a database. The web application lets administrators manage asset data, track asset history and create QR codes, from asset identification numbers. The created QR codes can be stuck to the physical assets. With the Android application administrators and students can scan an asset QR code to see that asset's specifications. The system also lets users borrow and return assets by scanning QR codes.

Strengths of the Existing System



The system offers strengths for institutional asset management. First the system replaces paper-based and spreadsheet-based records with a cloud database. Second the system uses QR codes so that assets can be identified quickly by scanning with a device. Third the system allows administrators to generate and print QR codes for labeling assets. Fourth the system records asset history and circulation so administrators can see who used an asset and when it was borrowed or returned. The system also supplies geolocation information for tracking assets, which helps in locating lost or misplaced equipment.

Another strength of the system is that the system offers interfaces, for different users. The web application is used for managing asset data and monitoring circulation while the Android application lets administrators and students scan QR codes and access asset information. The applications are connected to a database so the system allows information to be shared between the different platforms.

Weakness of the Existing System

Despite useful features the study points out several challenges when putting the system into use.

The system needs an Internet connection and good infrastructure. If Internet bandwidth is low the system may not work well. The study also notes that human resource readiness and ICT literacy are problems because staff and students are used to manual ways. The study says that training, supervision and technical support are needed to help people use the system.

The researchers also noted a drawback, in the geolocation feature. The system tracks assets only when users scan the QR code at the times. If users do not scan the QR code when needed the system may not record the movement of the asset correctly. Also the mobile application could run on Android smartphones during the study.

Research Gap

The study shows that using QR codes, cloud databases, mobile apps and asset tracking works well for schools. It offers features like registering assets creating QR codes scanning equipment, recording who uses each item viewing the history of an asset and tracking items through location data. These tools make a base for building an inventory system in educational settings.

However this system was mainly made for the Civil Engineering Department at Politeknik Negeri Balikpapan. It was built to manage types of TVET assets, both fixed and movable. The new system will focus on computer hardware, specifications and equipment inside an academic department. This narrower goal allows the system's fields and functions to be adapted specifically for computer-related items and the needs of the users.



The study, on Asset Guard: A Multi-Platform Asset Management System Using QR Code and Geolocation created an asset management system to help an organization keep track of and manage assets given to employees. The system was made to replace manual tasks, like asset encoding, inventory management report making and tracking where assets are located. It was designed to reduce issues that come with work especially when it comes to how fast things get done how reliable the information is and how secure the data is. The system was built using the Extreme Programming model. Used QR codes and geolocation to help with identifying and tracking assets.

Figure 4. Screenshot of the Asset Guard Asset Management System

The screenshot shows a web browser window with the URL 127.0.0.1/asset/supply.html?version=0.900968327108115#. The page title is "DIMMSU ASSET GUARD". It displays a table of assets with the following columns: Account Name, Article, Description, Property Number, Unit of Measure, Unit Value, Date of Acquisition, Issued to Employee, Issued to Department/Unit, Quantity, Status, and Action. There are four rows of asset data, each with a "PREVIEW" button and a "REMOVE" button in the Action column.

Account Name	Article	Description	Property Number	Unit of Measure	Unit Value	Date of Acquisition	Issued to Employee	Issued to Department/Unit	Quantity	Status	Action
0938-0965-7 Machinery	Grinder	(1) 4-1/2 In. Angle Grinder, (1) Grinding Wheel Guard, (1) Auxiliary Handle, (1) Inner (Clamping) Fl	0001-1	1	5000	November 13, 2019	Patacsil Joseph A	n/a	1	Condemned	PREVIEW REMOVE
0938-0965-7 Machinery	Table Saw	Moton: 15 amps, 120v. Blade Diameter: 10" Arbor Size: 5/8" Depth of Cut at 0": 3" Depth of Cut at 45	9998-1	1	67899	November 26, 2019	n/a	COT	1	Serviceable	PREVIEW REMOVE
0938-0965-7 Machinery	Jack Hammer	Impacts per minute: 1,450 ipm. Dimensions: 652 mm x 121 mm x 231 mm (25-3/4" x 4-3/4" x 9-1/8") Net	9865-2	1	89765	November 19, 2019	Rogers Steve G	n/a	1	Serviceable	PREVIEW REMOVE
0938-0965-7 Machinery	Wheelbarrow	Alibaba.com offers 146 wheel barrow specifications standard products. About 42% of these are wheelba	9876-1	1	4500	December 25, 2019	Calabading Jasper, M. Rogers Steve G	n/a	1	Serviceable	PREVIEW REMOVE

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The system includes modules to help manage organizational assets. The Encode Asset module lets users input asset details and record the asset's location using OpenStreetMap. The Manage Asset module allows users to update asset records change the status of assets, delete records and print QR codes. The QR sticker that is generated contains the asset information and a QR code. This QR code can be scanned using the application to get more information, about the asset.

Strengths of the Existing System

Asset Guard system is very helpful for keeping track of things. I notice Asset Guard system cuts down the need for paper records by turning asset details into data. Asset Guard system uses QR codes so a phone can read a tag and pull up all the info about an item. Asset Guard system lets people see whether equipment is working needs fixing or should be discarded. I see Asset Guard system can show the spot of an item, on a map using OpenStreetMap. Asset Guard system also prints out



inventory lists, card sheets and receipts automatically so people do not have to write the things many times.

Weakness of the Existing System

Asset Guard provides asset management functions. Its main purpose is to manage organizational assets that are issued to employees. The system focuses on tracking employee-issued property conducting inventory checks and recording details about company-owned equipment. Users are required to scan and update asset information regularly to ensure that records remain accurate and up to date.

In comparison to the proposed system Asset Guard does not focus on managing computer hardware or equipment inventory within a department. Its asset data is designed for organizational use not for detailed tracking of computer-specific items or equipment used in educational settings, like schools.

Research Gap

The Asset Guard study shows that QR codes can be used well in a multi-platform asset management system. The study lists functions such as asset encoding, QR-code generation QR-code scanning, status updating, location tracking, inventory management and report generation. These ideas help build inventory systems.

The original system was mainly for managing assets that employees use. The new system will focus on computer hardware and equipment in a college department. This lets the new system give inventory details that match computer gear, such, as equipment details, specifications, location and status.

So the Asset Guard study gives a base for using QR-code identification, asset status monitoring and digital inventory management in the system. The new system takes those ideas and tailors them for the needs of managing computer hardware and equipment in a school setting.

Automating Inventory for University Assets: A Software-Based Approach

Austin (2025) created a software-based automated inventory system to manage assets at Plateau State University. The study focused on the challenges caused by the university's manual and semi-manual asset management methods. These methods relied heavily on paper-based records and spreadsheets. This led to problems. Data was often recorded incorrectly. Retrieval of information was difficult and slow. Asset details were not easy to access. Processing times were long. There was no real-time visibility into asset movements. To fix these issues the researcher built a web-based system. The system was developed using PHP, MySQL and Tailwind CSS. It was designed to handle the asset lifecycle. This includes asset acquisition. It covers asset deployment. It supports asset maintenance. It manages asset disposal. The goal was to make asset management faster more accurate and more efficient.



Figure 5. Screenshot of the Automated Inventory System



The system gives institutions a way to manage their assets. It makes sure authorized people can log in. Access levels are set based on roles. All data stays in one place. You can see what is happening in time. Reports are easy to understand. Data is shown in ways. Every change is tracked. The system handles both things—like buildings and machines—and non-physical things—, like ideas and computer programs.

Strengths of the Existing System

The system has benefits for university asset management. First the system takes the place of paper and spreadsheet records by using a database. Second the system provides real time visibility of asset conditions, usage and availability through dashboards. Third the system keeps audit trails that increase accountability by recording every action, inside the system. Fourth the system uses automated workflows to reduce the time needed for asset registration, maintenance and disposal. The system also provides analytics and reporting tools that help the institution make decisions.



Another benefit of the system is its ability to manage the asset lifecycle. Of keeping track of inventory the system follows an asset from purchase and deployment to maintenance and final disposal.

Weakness of the Existing System

I notice that the system offers a way to handle university assets. The system includes many different kinds of assets. These include buildings, tools, ideas and programs. Because the system covers things the system is not made to focus on the specific details needed for computer parts and equipment.

The system is also, about keeping track of everything in one place managing how assets are used over time watching them making reports and keeping records. Unlike the system this study does not mainly use QR codes so people can quickly find computer parts and get their details by scanning.

Research Gap

I think Austins (2025) study shows how important it is to replace spreadsheets with a software inventory system in a university. Austins study highlights features such as records that live in one place real-time monitoring, detailed reports, secure login and tracking the life of an item. Austins study gives ideas that help design the system.

I believe the current system manages kinds of university assets of focusing on computer hardware and equipment. The new system will be built for one department concentrating on computer hardware and equipment. The new system will provide data such as hardware specifications, equipment details, location and status. The new system will also use QR-code scanning so users can get information quickly and it will produce QR codes when new items are added.

I see that Austins study gives a base for an automated inventory management, in a school while the new system adapts those ideas to the specific needs of computer hardware and equipment in an academic department.

Theoretical Framework



The development of the **QR Code-Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department** is anchored on theoretical perspectives that explain user acceptance of technology, the interaction of system components, and the improvement of operational processes. These foundations provide a basis for designing a system that can support the identification, monitoring, tracking, and management of computer laboratory equipment.

1. Technology Acceptance Model (TAM) - Davis (1989)

The **Technology Acceptance Model (TAM)** developed by Fred Davis explains how users accept and adopt information technologies. The model emphasizes two major factors: **perceived usefulness** and **perceived ease of use**. Perceived usefulness refers to the extent to which a person believes that using a particular system will improve work performance, while perceived ease of use refers to the extent to which a person believes that using the system will require minimal effort.

Application to the Study

TAM is relevant to the proposed QR Code Based Smart Inventory and Asset Tracking System because the effectiveness of the system depends partly on whether laboratory personnel and other authorized users can understand and use its functions. QR code scanning can provide a convenient method of identifying equipment, while functions such as inventory recording, equipment status monitoring, check-in and check-out tracking, maintenance recording, and report generation can improve the usefulness of the system. Applying the principles of perceived usefulness and perceived ease of use can guide the researchers in developing a system that is practical and accessible to its intended users.

2. General Systems Theory - von Bertalanffy (1968)



General Systems Theory, associated with Ludwig von Bertalanffy, views a system as a set of interconnected and interdependent components that interact with one another to achieve a common purpose. The theory emphasizes that individual components should not be viewed independently because their relationships and interactions contribute to the performance of the entire system.

Application to the Study

General Systems Theory is applicable to the proposed inventory and asset tracking system because the system consists of interconnected components, including users, computer laboratory equipment, QR codes, inventory records, databases, and administrative processes. The system receives equipment and user information as inputs, processes these through registration, QR code identification, inventory updating, transaction recording, and data management, and produces outputs such as updated inventory records, equipment status information, borrowing history, maintenance records, and reports. By connecting these components within one system, the proposed solution aims to improve the coordination and efficiency of computer laboratory asset management.

3. Lean Management Principles - Ohno (1988)

Lean Management Principles, associated with the work of Taiichi Ohno, emphasize the reduction of unnecessary activities, elimination of waste, improvement of workflow, and efficient use of resources. These principles focus on identifying activities that do not add value and improving processes to make operations more efficient and systematic.

Application to the Study



Lean Management Principles are relevant to the study because the existing manual inventory process may involve repetitive searching, manual encoding, verification of records, and updating of spreadsheets. These activities may consume time and increase the possibility of inconsistent or outdated information. The proposed QR-based system applies the principle of process improvement by reducing unnecessary manual encoding and simplifying equipment identification through QR code scanning. Centralized records can also reduce duplication and make equipment information easier to retrieve and update. Through these improvements, the proposed system aims to provide a more streamlined and efficient inventory process for the CEIT Department.

Synthesis of the Theoretical Framework

The three theoretical foundations complement one another in guiding the development of the proposed system. **Technology Acceptance Model** provides a basis for considering the usefulness and ease of use of the system from the perspective of its intended users. **General Systems Theory** provides a basis for understanding how the users, equipment, QR codes, database, and inventory processes interact as interconnected components of one system. **Lean Management Principles** provide a basis for improving the efficiency of the existing inventory process by reducing unnecessary manual activities and streamlining asset-management procedures.

Together, these foundations guide the researchers in developing a QR-based inventory and asset tracking system that is user-oriented, interconnected, and focused on improving the efficiency of computer laboratory asset management.



Conceptual Framework

(Existing Conceptual Framework)

The existing conceptual framework describes the current inventory management process used by the College of Engineering and Information Technology (CEIT) Department for monitoring and managing computer laboratory equipment. Based on the existing inventory records, the



department uses a spreadsheet-based manual inventory system to record information about its laboratory properties and defective equipment. The framework follows the Input-Process-Output (IPO) Model) to illustrate how equipment information is collected, manually recorded and monitored, and converted into inventory and defective-item reports.

Input

The input phase consists of the equipment and inventory information collected by the CEIT Department. This includes the ID number, item name, quantity, unit, classification, description, year encoded, known acquisition date, remarks, checked status, last date checked, and inventory code of each laboratory asset. The inventory covers different types of equipment and materials, including computer units, Mac computers, laptops, adapters, cables, networking equipment, peripherals, and other laboratory tools.

The existing records also include information about defective equipment. The Defective Items Report contains the quantity, unit, item description, category, identified problem or test result, and date encoded. This information serves as the basis for identifying equipment that requires repair, replacement, inspection, or further action.

Process

The process phase involves the manual recording, updating, checking, and monitoring of laboratory equipment using a spreadsheet. Personnel enter the information of each asset into the inventory spreadsheet and assign an inventory code for identification. Equipment information is then manually updated whenever an item is checked, inspected, transferred, repaired, or identified as defective.



The inventory also requires personnel to manually check the condition and status of equipment. Statuses such as OK, Working, Defective, and Deployed are recorded in the spreadsheet, together with the date when the equipment was last checked. When defective equipment is identified, the information is separately recorded in the Defective Items Report, including the problem observed and the date it was documented.

This process depends heavily on manual data entry, checking, updating, and searching of spreadsheet records. Personnel must review the records to determine the current status and condition of equipment. As a result, inventory monitoring requires continuous manual effort and depends on the accuracy and timeliness of personnel when updating the records.

Output

The output of the existing inventory process consists of spreadsheet-based inventory records and defective equipment reports. These records provide information regarding the quantity, classification, condition, status, location-related information, and identification of laboratory equipment.

The existing process allows the CEIT Department to maintain a record of its available properties and identify equipment that is defective or requires attention. However, because the process is primarily manual and spreadsheet-based, it may have limitations in terms of real-time tracking, immediate updating, equipment movement monitoring, borrowing and return tracking, and centralized maintenance monitoring.

Therefore, the existing conceptual framework establishes the current condition of inventory management and provides the basis for identifying the limitations and gaps that the proposed QR



Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in CEIT Department aims to address.

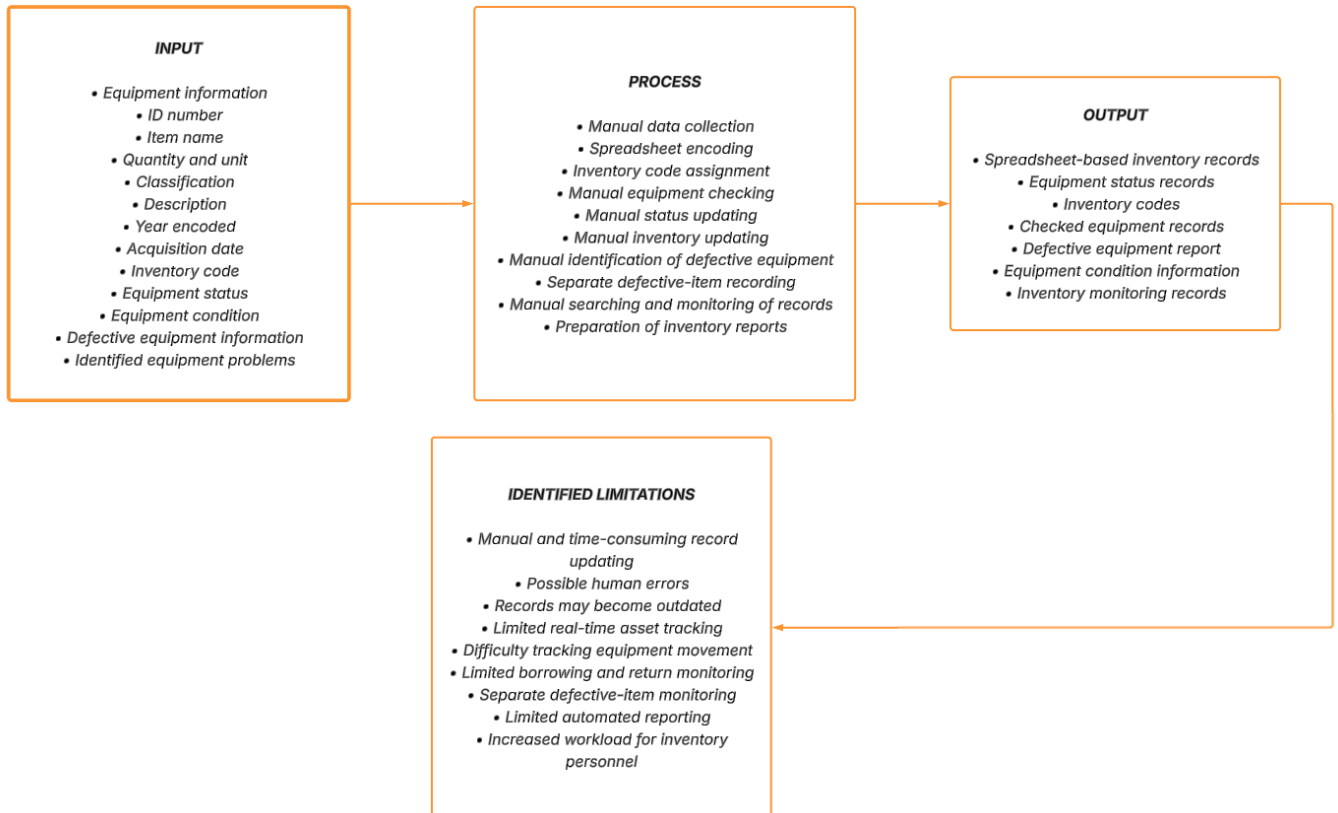


Figure : IPO Diagram for the flow of the Existing System

(Proposed Conceptual Framework)

The conceptual framework of this study is guided by the Input-Process-Output (IPO) Model, which explains how the proposed QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in CEIT Department transforms collected equipment and



inventory information into an organized, efficient, and reliable asset management system. The IPO model serves as the foundation for identifying the necessary inputs, development processes, and expected outputs of the study.

Input

The input phase of the study consists of the data, information, and requirements needed for the development of the proposed system. This includes information about the existing inventory practices of the CEIT Department and the computer laboratory equipment being managed, such as computer units, monitors, keyboards, mice, headsets, networking devices, and other laboratory assets. Data regarding equipment identification, location, condition, availability, borrowing records, and maintenance status are also considered.

Problem identification is conducted to determine the challenges encountered in the existing manual inventory process, including inaccurate or outdated records, difficulty in tracking borrowed or misplaced equipment, inefficient maintenance monitoring, and the time required to manually update inventory information. User requirements are then identified based on the needs of laboratory personnel and other authorized users who will interact with the system.

In addition, related literature and existing studies on QR code technology, inventory management, asset tracking, and digital laboratory management are reviewed to provide a theoretical and technical foundation for the proposed system. These inputs serve as the basis for determining the system requirements, features, and functionalities needed to address the identified problems.

Process



The process phase involves the Software Development Life Cycle (SDLC) using the Waterfall Model as the development approach for the proposed system. The process begins with the requirements analysis phase, where the researchers identify and document the functional and non-functional requirements of the system based on the existing inventory process and the needs of the CEIT Department.

This is followed by the system design phase, where the overall structure, database, user interface, QR code functionality, inventory management features, and asset-tracking processes are planned. The design includes features such as equipment registration, QR code generation and scanning, check-in and check-out monitoring, equipment status and condition tracking, maintenance records, user management, and inventory reporting.

During the development phase, the planned system components and functionalities are implemented into a working system. The QR codes are assigned to registered laboratory equipment, while the database is developed to store and manage asset information, borrowing transactions, maintenance records, and other relevant data.

The developed system then undergoes the testing phase to identify errors, bugs, functionality issues, and other problems that may affect system performance. The system is evaluated to determine whether its features function according to the identified requirements. After testing and necessary revisions, the system proceeds to the deployment phase, where it is prepared for implementation within the CEIT Department. The final phase involves maintenance, which includes monitoring the system and making necessary improvements or corrections to maintain its reliability and functionality.

Output



The output of the conceptual framework is a functional QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in CEIT Department. The system is expected to provide a centralized digital inventory that allows authorized personnel to efficiently identify, monitor, and manage laboratory equipment through QR code technology.

The expected outputs include accurate and organized inventory records, efficient equipment check-in and check-out monitoring, improved tracking of asset location and condition, better maintenance documentation, increased equipment accountability, reduced manual recording errors, and automated inventory reports. Ultimately, the system is expected to provide the CEIT Department with a more efficient, reliable, and organized approach to managing its computer laboratory equipment.

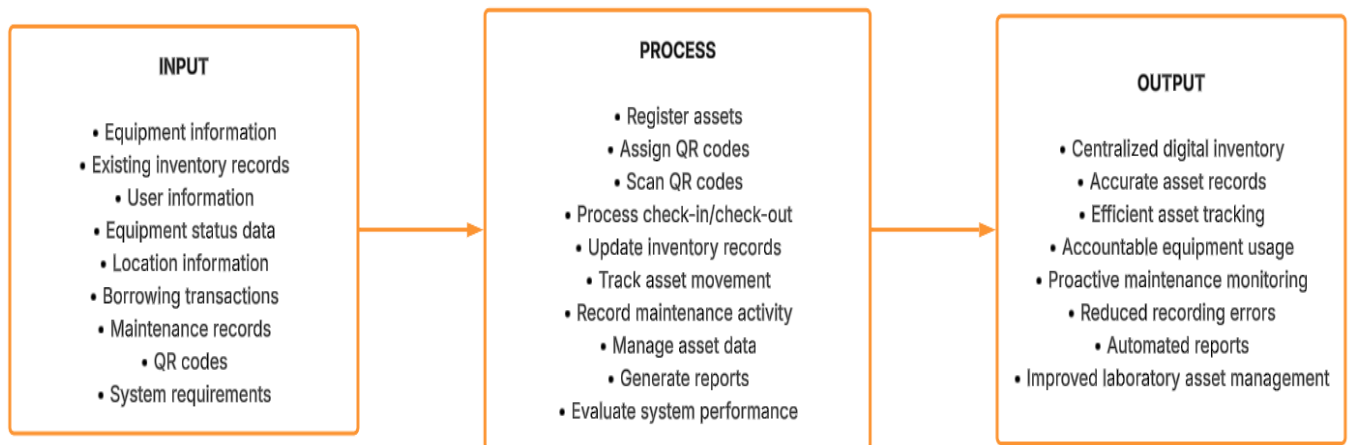


Figure : IPO Diagram for the flow of the Proposed System

Research Questions

This study aims to answer the following research questions:



1. What are the existing challenges encountered by the CEIT Department in managing computer laboratory equipment using the current manual and spreadsheet-based inventory process, particularly in terms of:
 - 1.1. accuracy and updating of inventory records;
 - 1.2. equipment identification and location tracking;
 - 1.3. borrowing and return monitoring;
 - 1.4. equipment condition and status monitoring; and
 - 1.5. maintenance and accountability of laboratory equipment?

2. What functional and non-functional requirements are needed for the development of a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department?

3. How can the proposed system be designed and developed to provide the following functionalities:
 - 3.1. equipment registration and inventory management;
 - 3.2. QR code generation and scanning;
 - 3.3. equipment check-in and check-out monitoring;
 - 3.4. asset location, status, and condition tracking;
 - 3.5. maintenance record management;



- 3.6. user accountability; and
 - 3.7. inventory report generation?
4. How can the developed system be implemented using the Software Development Life Cycle (SDLC) and Waterfall Model to address the identified inventory management requirements of the CEIT Department?
5. How effective is the developed QR Code Based Smart Inventory and Asset Tracking System in improving the management of computer laboratory equipment in terms of:
- 5.1. functionality;
 - 5.2. usability;
 - 5.3. efficiency;
 - 5.4. reliability; and
 - 5.5. asset accountability?
6. To what extent does the developed system address the limitations of the existing manual and spreadsheet-based inventory process, particularly in reducing manual recording, searching, updating, and monitoring of computer laboratory equipment?

Research Objectives

General Objective



The general objective of this study is to design and develop a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department that will improve the efficiency, accuracy, monitoring, maintenance, and accountability of computer laboratory equipment through centralized digital inventory management and QR code-based asset identification.

Specific Objectives

Specifically, this study aims to:

1. Identify and analyze the existing inventory management practices and challenges encountered by the CEIT Department in managing computer laboratory equipment, particularly in terms of inaccurate or outdated records, equipment misplacement, borrowing and return monitoring, equipment condition, asset location, and maintenance tracking.
2. Determine the functional and non-functional requirements of the proposed QR Code Based Smart Inventory and Asset Tracking System based on the existing inventory process and the needs of authorized laboratory personnel.
3. Design and develop a QR code-based smart inventory and asset tracking system that provides centralized equipment records and efficient identification, monitoring, and updating of laboratory assets.



4. Integrate essential system functionalities, including equipment registration, QR code generation and scanning, check-in and check-out monitoring, asset status and condition tracking, location monitoring, maintenance records, user accountability, and inventory reporting.
5. Implement the proposed system using the Software Development Life Cycle (SDLC) and Waterfall Model to ensure systematic requirements analysis, system design, development, testing, deployment, and maintenance.
6. Evaluate the developed system in terms of functionality, usability, efficiency, reliability, and its ability to improve inventory management, asset accountability, equipment monitoring, and maintenance documentation within the CEIT Department.
7. Determine whether the developed system can reduce the limitations associated with the existing spreadsheet-based inventory process, particularly manual recording, searching, updating, and monitoring of laboratory equipment.

Significance of the Study

The study entitled “**Design and Development of a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in CEIT Department**” is significant because it seeks to address the challenges associated with the existing manual and spreadsheet-based inventory process used for managing computer laboratory equipment. By



integrating QR code technology with centralized digital inventory management, the proposed system is intended to improve equipment identification, monitoring, accountability, maintenance documentation, and accessibility of inventory information.

The findings and developed system may provide benefits to the following:

College of Engineering and Information Technology (CEIT) Department

The **College of Engineering and Information Technology (CEIT) Department** may benefit from a more organized and centralized approach to managing its computer laboratory equipment. The proposed system can help the department maintain updated records of laboratory assets, monitor their status and condition, track equipment movement, document maintenance activities, and improve equipment accountability. This may support more efficient laboratory operations and better utilization of the department's resources.

Laboratory Personnel and Administrators

Laboratory personnel and administrators may benefit from reduced reliance on manual spreadsheet-based inventory procedures. QR code scanning can provide a faster means of identifying equipment, while centralized records can make information regarding asset status, location, borrowing transactions, and maintenance history easier to retrieve and update. The system may also assist personnel in monitoring equipment that requires inspection, repair, or other corrective action.

Faculty Members

Faculty members may benefit from improved availability and monitoring of laboratory resources. More accurate inventory records can help faculty members determine the availability and



condition of equipment needed for laboratory activities and academic instruction. The improved tracking of equipment may also support greater accountability for assets used during academic activities.

Students

Students may indirectly benefit from better-managed and properly maintained computer laboratory equipment. Improved monitoring and timely identification of defective or unavailable equipment may contribute to a more organized laboratory environment and help ensure that necessary resources are available for academic activities.

Siena College of Taytay

Siena College of Taytay may benefit from the study through the application of a technology-based solution to an actual institutional inventory-management concern. The proposed system may demonstrate how QR code technology, database management, and software development can be applied to improve the management and accountability of institutional resources. It may also serve as a reference for future digitalization initiatives within the institution.

Future Researchers

Future researchers may use this study as a reference for related studies involving QR code technology, inventory management, asset tracking, laboratory management systems, and other digital solutions for educational institutions. The study may also provide a foundation for future improvements, additional features, or similar systems designed for other departments or organizations.



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Scope and Delimitations

Scope of the Study

This study focuses on the **design and development of a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department.**



The proposed system is intended to improve the existing manual and spreadsheet-based inventory process by providing a centralized digital platform for recording, identifying, monitoring, and managing computer laboratory equipment.

The study covers the computer laboratory equipment managed by the CEIT Department, including computer units, monitors, keyboards, mice, headsets, networking devices, cables, adapters, and other relevant laboratory assets. The system will maintain important asset information such as equipment identification, item description, classification, location, status, condition, availability, and other relevant inventory details.

The proposed system will incorporate **QR code technology** to provide unique identification for registered laboratory equipment. Authorized users will be able to generate and scan QR codes to retrieve and update corresponding equipment information. The system will also include inventory management functions such as equipment registration, QR code generation and scanning, equipment check-in and check-out, asset status and condition monitoring, asset location tracking, maintenance record management, user accountability, and inventory report generation.

The study also covers the development of a centralized database for storing and managing equipment information, borrowing and return transactions, maintenance records, and other relevant asset-management data. The system will be developed using the **Software Development Life Cycle (SDLC) with the Waterfall Model**, covering requirements analysis, system design, development, testing, deployment, and maintenance.

The developed system will be evaluated based on selected software quality and user-related criteria, particularly its functionality, usability, efficiency, and reliability. The evaluation will focus on determining whether the proposed system can address the identified weaknesses of the existing



inventory process and improve the management and accountability of CEIT computer laboratory equipment.

Delimitations of the Study

This study is limited to the inventory and asset management of **computer laboratory equipment within the CEIT Department**. It does not cover inventory management for other departments, offices, classrooms, or institutional facilities outside the defined scope of the study.

The system is primarily intended for **authorized CEIT personnel and designated users** responsible for managing or monitoring laboratory equipment. It is not designed as a general-purpose inventory system for external organizations or unrestricted public use.

The study focuses on equipment identification, inventory recording, asset tracking, borrowing and return monitoring, condition and status monitoring, maintenance documentation, user accountability, and report generation. It does not include automated physical tracking technologies such as GPS, RFID, Bluetooth-based tracking, or real-time location tracking. QR codes are used primarily for equipment identification and retrieval of associated digital records.

The proposed system does not automatically determine whether a computer has unauthorized software, games, or other applications installed. Although the background of the study recognizes unauthorized software installation as one of the concerns related to laboratory computer management, the proposed system is limited to recording and monitoring equipment information rather than performing automated computer-system auditing or software detection.

The system also does not include automated repair of defective equipment, procurement of replacement equipment, financial accounting, depreciation computation, or automated purchasing



processes. Maintenance functionality is limited to recording and monitoring maintenance-related information.

The study is also delimited by the data and information made available by the CEIT Department. The accuracy and completeness of inventory records generated by the system depend on the information entered and maintained by authorized users. The effectiveness of the system may therefore be influenced by proper user adoption, accurate data entry, availability of the required devices for QR scanning, and compliance with established inventory procedures.

Finally, the study focuses on the development and evaluation of the proposed system within the defined research setting and does not claim that the system is universally applicable to all educational institutions. Future implementation in other departments or organizations may require modifications based on their specific inventory procedures, equipment types, user requirements, and operational conditions.

Definition of Terms

This section presents the important terms used in the study **“Design and Development of a QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in CEIT Department.”** The terms are defined according to their operational meaning within the study or their contextual meaning based on their accepted use in information technology,



inventory management, and software development. The terms are arranged alphabetically to provide consistency and ease of reference.

Asset.

Refers to any computer laboratory equipment or resource owned, managed, or used by the CEIT Department that is recorded and monitored within the proposed system.

Asset Accountability.

Refers to the process of identifying and recording the users responsible for the use, borrowing, transfer, return, or handling of laboratory equipment.

Asset Location Tracking.

Refers to the monitoring and recording of the current or assigned location of computer laboratory equipment within the CEIT Department.

Asset Tracking.

Refers to the process of monitoring the identification, location, status, condition, movement, and usage history of laboratory equipment through the proposed system.

Check-in.

Refers to the process of recording the return of borrowed laboratory equipment to the CEIT Department or its designated location.

Check-out.



Refers to the process of recording the release or borrowing of laboratory equipment by an authorized user.

Computer Laboratory Equipment.

Refers to computer units, monitors, keyboards, mice, headsets, networking devices, cables, adapters, and other technological resources used or maintained in the CEIT Department's computer laboratories.

Condition.

Refers to the physical or operational state of a laboratory asset as recorded in the proposed system, such as working, defective, damaged, or other applicable conditions.

CEIT Department.

Refers to the **College of Engineering and Information Technology Department** of Siena College of Taytay, which serves as the research setting and intended institutional environment for the proposed inventory and asset tracking system.

Centralized Database.

Refers to the organized digital storage of equipment, user, transaction, location, condition, and maintenance information that can be accessed and managed through the proposed system.

Defective Equipment.

Refers to laboratory equipment identified as malfunctioning, damaged, or otherwise unable to perform its intended function and recorded as requiring inspection, repair, replacement, or other appropriate action.



Equipment Inventory.

Refers to the systematic record of computer laboratory equipment maintained by the CEIT Department, including information such as identification, description, quantity, location, condition, status, and other relevant asset details.

Inventory Management.

Refers to the process of recording, organizing, monitoring, updating, and maintaining information about computer laboratory equipment within the CEIT Department.

Inventory Report.

Refers to a generated summary of equipment information from the proposed system, which may include asset identification, quantity, location, condition, status, borrowing records, and maintenance information.

Maintenance Record.

Refers to the documented information concerning the inspection, repair, servicing, maintenance activity, or other corrective action performed on laboratory equipment.

QR Code.

Refers to a two-dimensional machine-readable code assigned to a laboratory asset in the proposed system. When scanned, it provides access to the corresponding equipment record stored in the system.

QR Code Scanning.



Refers to the process of using a compatible device or scanner to read the QR code assigned to a laboratory asset and retrieve its corresponding information from the system.

Smart Inventory System.

Refers to the proposed digital system that combines centralized inventory management, QR code identification, asset tracking, transaction monitoring, condition monitoring, maintenance documentation, and reporting for CEIT computer laboratory equipment.

Software Development Life Cycle (SDLC).

Refers to the systematic process used in the study for planning, analyzing, designing, developing, testing, deploying, and maintaining the proposed inventory and asset tracking system.

Status.

Refers to the current operational or inventory state of an equipment asset as recorded in the proposed system, such as available, borrowed, defective, under maintenance, or deployed.

User Accountability.

Refers to the system-supported recording of information that identifies authorized users associated with the borrowing, return, handling, or use of laboratory equipment.

Waterfall Model.

Refers to the sequential software development approach adopted in the study in which requirements analysis, system design, development, testing, deployment, and maintenance are performed as structured development phases.



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CHAPTER 2

Methods

Research Study



Research Design

This study employs a **developmental research design** because it focuses on the systematic design, development, implementation, and evaluation of a **QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department**. The study is intended to address the identified limitations of the existing manual and spreadsheet-based inventory process by developing a technology-based solution that can improve the management, monitoring, tracking, and accountability of computer laboratory equipment.

Developmental research design is appropriate for this study because it involves the systematic process of identifying existing problems, determining user and system requirements, designing the proposed solution, developing its functionalities, testing the system, and evaluating its effectiveness. The researchers will first examine the existing inventory practices of the CEIT Department to identify challenges related to equipment identification, inventory record accuracy, borrowing and return monitoring, asset location, equipment condition, maintenance tracking, and accountability.

The study will use the **Software Development Life Cycle (SDLC) through the Waterfall Model** as the development approach. The development process will consist of requirements analysis, system design, development, testing, deployment, and maintenance. The proposed system will incorporate features such as equipment registration, QR code generation and scanning, centralized inventory records, check-in and check-out monitoring, asset status and condition tracking, maintenance records, user accountability, and inventory report generation.



After development, the system will be tested and evaluated to determine whether its functionalities operate according to the identified requirements and whether it can improve the existing inventory management process. The evaluation will consider relevant criteria such as functionality, usability, efficiency, reliability, and asset accountability.

Through this research design, the study seeks not only to develop a functional system but also to determine its suitability in addressing the specific inventory and asset-management needs of the CEIT Department. The developmental approach therefore provides a structured basis for transforming the identified problems in the existing inventory process into a practical and technology-based solution.

Research Instruments

The researchers will use structured research instruments to gather the necessary data for the design, development, and evaluation of the **QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department**. The instruments will



be used to understand the existing inventory management process, identify problems encountered by users, determine system requirements, and evaluate the developed system.

The primary research instrument will be a **structured questionnaire** distributed to selected respondents who have direct experience with or interaction with the computer laboratory equipment and inventory process. The respondents may include laboratory personnel, faculty members, and other authorized users of the computer laboratories. The questionnaire will consist primarily of closed-ended questions and rating-scale items designed to gather information regarding the existing inventory process, equipment monitoring, borrowing and return procedures, maintenance tracking, asset accountability, and the perceived usefulness of the proposed system.

The questionnaire will also include items that assess the developed system based on relevant criteria such as **functionality, usability, efficiency, reliability, and asset accountability**. This will allow the researchers to determine whether the proposed system meets the identified requirements and improves the existing inventory management process.

In addition to the questionnaire, **structured interviews** may be conducted with selected CEIT laboratory personnel or administrators who are directly involved in managing computer laboratory equipment. The interviews will be used to obtain more detailed information about the existing inventory procedures, problems encountered in maintaining equipment records, and specific requirements that may not be fully captured through the questionnaire.

The researchers will also use **documentary observation and existing inventory records** as supporting sources of information. Existing spreadsheet-based inventory records, defective equipment records, and related inventory documents will be examined to understand the current data structure, equipment classifications, asset information, and existing monitoring practices.



These records will serve as references in identifying the data fields and functionalities required for the proposed system.

The combination of questionnaires, interviews, observation, and existing inventory records will provide the researchers with sufficient information to identify the current problems, establish system requirements, and evaluate the developed QR code-based inventory and asset tracking system.

Data Gathering

The data gathering process will be conducted systematically to obtain accurate and relevant information needed for the design, development, and evaluation of the **QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department**.



First, the researchers will coordinate with the **CEIT Department** and identify the appropriate respondents and personnel who are directly involved in the management, monitoring, and use of computer laboratory equipment. The researchers will also identify and examine the existing inventory records and related documents used by the department. These records will provide information about the types of equipment, inventory codes, quantities, conditions, statuses, and other relevant asset information.

Second, the researchers will conduct **observation and interviews** with selected laboratory personnel and administrators to understand the existing inventory procedures. The researchers will document how equipment is currently registered, checked, monitored, borrowed, returned, inspected, and recorded. The interviews will also be used to identify problems encountered during the current manual and spreadsheet-based inventory process and to determine the requirements of the proposed system.

Third, a **structured questionnaire** will be distributed to the selected respondents. The questionnaire will gather information regarding the existing inventory process, common problems encountered, user needs, and expectations for the proposed system. For the evaluation of the developed system, the questionnaire will also be used to obtain respondents' assessments of the system based on the identified evaluation criteria, such as functionality, usability, efficiency, reliability, and asset accountability.

Fourth, the researchers will collect, organize, and review the gathered data. Responses from the questionnaires will be tabulated and analyzed, while information obtained from interviews, observations, and existing inventory records will be examined to identify recurring problems, requirements, and necessary system functionalities.



Finally, the analyzed data will serve as the basis for the **requirements analysis and system design** of the proposed system. The identified requirements will guide the development of features such as equipment registration, QR code generation and scanning, check-in and check-out monitoring, asset status and condition tracking, maintenance records, user accountability, and inventory reporting. The gathered data will also serve as a basis for evaluating whether the developed system effectively addresses the limitations of the existing inventory management process in the CEIT Department.

Research Development

Software Development Methodology

This study will utilize the **Waterfall Model** as the software development methodology for the proposed **QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department**. The Waterfall Model is a structured and



sequential software development approach in which each phase is completed before proceeding to the next phase. It is appropriate for the study because the researchers can identify and document the system requirements before proceeding with the design and development of the proposed system.

The development process will consist of the following phases:

1. Requirements Analysis

In this phase, the researchers will gather, document, and analyze the requirements of the proposed system based on the data obtained from existing inventory records, observations, interviews, and questionnaires. The researchers will identify the problems and limitations of the current manual and spreadsheet-based inventory process and determine the functional and non-functional requirements of the system.

The requirements will include essential functionalities such as equipment registration, QR code generation and scanning, inventory management, check-in and check-out monitoring, asset status and condition tracking, location monitoring, maintenance record management, user accountability, and inventory report generation.

2. System Design

In this phase, the researchers will develop the overall design of the proposed system based on the identified requirements. The system architecture, database structure, user interface, navigation, and data flow will be planned and documented.

Relevant system models and diagrams, such as the **Data Flow Diagram (DFD)**, **Entity-Relationship Diagram (ERD)**, **system architecture diagram**, and **database design**, will be developed to illustrate the relationship between users, system processes, equipment records,



transactions, and stored data. The QR code identification and asset-tracking processes will also be incorporated into the system design.

3. Implementation and Development

In this phase, the researchers will translate the approved system design into a functional software application. The database, user interface, system modules, and required functionalities will be developed and integrated.

The implementation will include equipment registration, QR code generation and scanning, inventory record management, check-in and check-out transactions, asset status and condition monitoring, location tracking, maintenance records, user management, and inventory reporting. The developed components will be integrated to ensure that the system operates as a unified inventory and asset-tracking platform.

4. Testing

In this phase, the developed system will undergo systematic testing to identify errors, bugs, inconsistencies, and functionality issues. The researchers will verify whether each system feature performs according to the identified requirements.

Testing will include checking the functionality of QR code generation and scanning, equipment registration, inventory updating, check-in and check-out transactions, status and condition updates, maintenance records, user accountability, and report generation. Identified errors and issues will be corrected and retested to improve the overall reliability and functionality of the system.

5. Deployment and Maintenance



After the system has successfully completed the testing phase, the developed system will be prepared for implementation within the defined scope of the CEIT Department. The researchers will ensure that the system is properly configured and that the necessary users are provided with appropriate access.

The maintenance phase will involve monitoring the system for errors or issues discovered after implementation and applying necessary corrections or improvements. Maintenance may also include updating system components, correcting data-related issues, and improving existing functionalities when necessary.

Rationale for Using the Waterfall Model

The Waterfall Model is selected because it provides a **clear, systematic, and well-documented development process** that is appropriate for the scope and objectives of the study. The sequential structure allows the researchers to establish the system requirements before proceeding to design, development, testing, and deployment. This approach also supports proper documentation and evaluation of each development phase, which is important in an academic system-development study.

Through the Waterfall Model, the researchers can systematically transform the identified limitations of the existing CEIT inventory process into a functional QR code-based inventory and asset tracking system designed specifically for the management of computer laboratory equipment.



Current System

The current inventory management process of the **College of Engineering and Information Technology (CEIT) Department** is primarily manual and spreadsheet-based. Records of computer laboratory equipment are maintained through documents and spreadsheet files, such as Microsoft Excel. These records contain information including the item name, quantity, classification, condition, status, location, inventory code, and other relevant asset details.



The monitoring of equipment also involves manual checking and updating of inventory records. When equipment is borrowed, returned, transferred, inspected, or identified as defective, laboratory personnel are required to manually update the corresponding records. This process requires continuous attention and depends on the accuracy and timeliness of personnel when recording information.

The current process also presents difficulties in monitoring equipment movement, borrowing and return transactions, condition, and maintenance activities. Locating a particular item or determining its current status may require personnel to search through spreadsheet records or physically inspect the laboratory. Similarly, identifying defective equipment and reviewing its maintenance history may require checking separate records.

Furthermore, the existing process does not provide an integrated mechanism for QR-based equipment identification, centralized asset tracking, automated transaction recording, or systematic report generation. These limitations may result in outdated or inconsistent records, increased manual workload, difficulty in monitoring equipment, and challenges in maintaining asset accountability.

Proposed System

The proposed **QR Code Based Smart Inventory and Asset Tracking System for Computer Laboratory Equipment in the CEIT Department** is designed to improve the existing inventory management process by providing a centralized digital platform for recording, identifying, monitoring, and managing computer laboratory equipment.



Each registered laboratory asset will be assigned a unique QR code that corresponds to its digital inventory record. Authorized users can scan the QR code using a compatible device to quickly retrieve relevant information about the equipment, such as its identification, description, location, condition, status, availability, and transaction or maintenance history.

The proposed system will allow authorized users to perform the following functions:

- Register and manage laboratory equipment;
- Generate a unique QR code for each registered asset;
- Scan QR codes to retrieve equipment information;
- Update equipment information and status;
- Monitor equipment location and movement;
- Record equipment check-in and check-out transactions;
- Monitor equipment condition and availability;



- Record equipment maintenance and repair activities;
- Manage user accounts and user roles;
- Search and filter inventory records;
- Generate inventory and asset reports; and
- Maintain organized and centralized equipment records.

Through these functionalities, the proposed system aims to reduce dependence on manual spreadsheet-based inventory management, minimize recording errors, improve accessibility of asset information, strengthen equipment accountability, and support more efficient monitoring and maintenance of computer laboratory equipment.

Functional Requirements

The proposed system shall be able to perform the following functions:

1. User Authentication and Access Control



- The system shall allow authorized users to log in using registered accounts.
- The system shall provide role-based access according to the user's assigned role.
- The system shall support appropriate user roles such as **Administrator, Faculty, Laboratory Staff, and Student.**
- The system shall restrict system functions according to the permissions assigned to each user role.

2. Item Registration and Management

- The system shall allow authorized administrators or laboratory personnel to add new inventory items.
- The system shall allow authorized users to edit and update equipment information.
- The system shall allow authorized users to remove or archive inventory records when necessary.



- The system shall record relevant equipment information, including item name, description, category, inventory code, condition, status, location, and other applicable details.

3. QR Code Generation

- The system shall generate a unique QR code for each registered inventory item.
- The system shall associate each QR code with its corresponding equipment record.
- The system shall allow authorized users to view and print QR codes for equipment identification.

4. QR Code Scanning

- The system shall allow authorized users to scan the QR code assigned to an equipment item using a compatible camera-enabled device.
- The system shall retrieve the corresponding equipment record after a successful scan.
- The system shall display relevant equipment information after scanning.



- The system shall allow authorized users to update applicable equipment information through the retrieved record.

5. Inventory and Asset Tracking

- The system shall maintain centralized records of laboratory equipment.
- The system shall allow authorized users to update the current status of equipment.
- The system shall record equipment location information.
- The system shall monitor equipment availability.
- The system shall maintain relevant asset movement and transaction history.

6. Check-In and Check-Out Management

- The system shall allow authorized users to record the borrowing of equipment.
- The system shall record the borrower, date, time, and relevant transaction information.



- The system shall allow authorized personnel to record returned equipment.
- The system shall update the equipment's availability and status based on completed transactions.
- The system shall maintain a history of borrowing and return transactions.

7. Equipment Condition and Maintenance Management

- The system shall allow authorized personnel to record the current condition of equipment.
- The system shall allow personnel to identify equipment requiring inspection or maintenance.
- The system shall record maintenance, repair, and inspection information.
- The system shall maintain the maintenance history of equipment.
- The system shall allow authorized users to update the equipment status after maintenance or repair.



8. Search and Filter

- The system shall allow users to search for inventory items.
- The system shall allow users to filter equipment based on categories, status, condition, location, and availability.
- The system shall provide multiple search criteria to facilitate faster retrieval of equipment information.

9. Report Generation

- The system shall generate inventory reports.
- The system shall generate equipment status and condition reports.
- The system shall generate borrowing and return records.
- The system shall generate maintenance records and equipment history reports.



- The system shall provide options for exporting applicable reports in supported formats.

Non-Functional Requirements

The proposed system shall satisfy the following quality requirements:

1. Usability

- The system shall provide a user-friendly and understandable interface.
- The system shall provide clear navigation and instructions.
- The system shall provide appropriate feedback after user actions.
- The system shall be responsive across supported desktop and mobile devices.

2. Performance

- The system shall respond to user requests within an acceptable period under normal operating conditions.
- QR code scanning and retrieval of equipment information shall be performed efficiently.



- The system shall support multiple authorized users without significant performance degradation under the expected operating conditions.

3. Reliability

- The system shall maintain accurate and consistent inventory records.
- The system shall validate important user inputs before storing information.
- The system shall minimize data-entry and transaction errors.
- The system shall provide appropriate error handling when system operations fail.

4. Security

- The system shall protect user accounts through secure authentication.
- The system shall implement role-based access control.
- The system shall restrict unauthorized access to inventory and user information.



- The system shall protect stored data from unauthorized modification or deletion.

5. Scalability

- The system shall be designed to accommodate additional laboratory equipment and inventory records.
- The system architecture shall allow additional features and functionalities to be incorporated in future system improvements.

6. Maintainability

- The system shall be structured to facilitate troubleshooting, modification, and future enhancement.
- The system source code shall follow consistent coding practices.
- System documentation shall be maintained to support future maintenance and development.

7. Compatibility

- The system shall be accessible through supported desktop and mobile devices.



- The system shall be compatible with commonly used modern web browsers, such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, subject to the technologies used in the final implementation.

Architectural Framework

The proposed **QR Code Based Smart Inventory and Asset Tracking System** will follow the **Model-View-Controller (MVC) architectural pattern**. MVC separates the application into three major components: the **Model**, **View**, and **Controller**. This separation allows the system's data management, user interface, and application logic to be organized independently, making the system easier to develop, maintain, test, and enhance.

Model (Data Layer)

The **Model** represents the data and business rules of the system. It manages the creation, retrieval, updating, and storage of information in the database.

The Model includes:

- User account and authentication information;
- User roles and permissions;
- Laboratory equipment records;



- QR code information associated with equipment;
- Equipment categories and classifications;
- Equipment status and condition;
- Equipment location information;
- Check-in and check-out transactions;
- Asset movement history;
- Maintenance and repair records; and
- Inventory and report-related data.

The Model is responsible for data validation, database operations, and maintaining the integrity and consistency of stored information.



View (Presentation Layer)

The **View** represents the user interface of the system. It provides the screens and interface elements through which authorized users interact with the application.

The View is responsible for:

- Displaying inventory records and equipment information;
- Providing login and account interfaces;
- Providing equipment registration forms;
- Displaying QR code information;
- Providing QR code scanning interfaces;
- Displaying equipment status, condition, and location;
- Providing check-in and check-out interfaces;



- Displaying maintenance information;
- Providing search and filtering functions;
- Displaying generated reports; and
- Providing feedback regarding completed or unsuccessful user actions.

The View will be designed to provide clear navigation and responsive interaction across supported devices.

Controller (Application Logic Layer)

The **Controller** serves as the intermediary between the View and Model. It receives user requests, processes system operations, applies business rules, and communicates with the Model to retrieve or modify data.

The Controller is responsible for:

- Processing user authentication requests;



- Validating user inputs;
- Enforcing role-based access permissions;
- Processing equipment registration and updates;
- Managing QR code generation and scanning requests;
- Processing check-in and check-out transactions;
- Updating equipment status, condition, and location;
- Recording maintenance activities;
- Retrieving inventory information;
- Processing search and filtering requests;



- Generating reports; and
- Handling errors and exceptions.

Through the interaction of the **Model, View, and Controller**, the proposed system will provide a structured architecture for managing laboratory equipment information, processing asset transactions, and presenting inventory information to authorized users.

Diagrams

Use Case Diagram

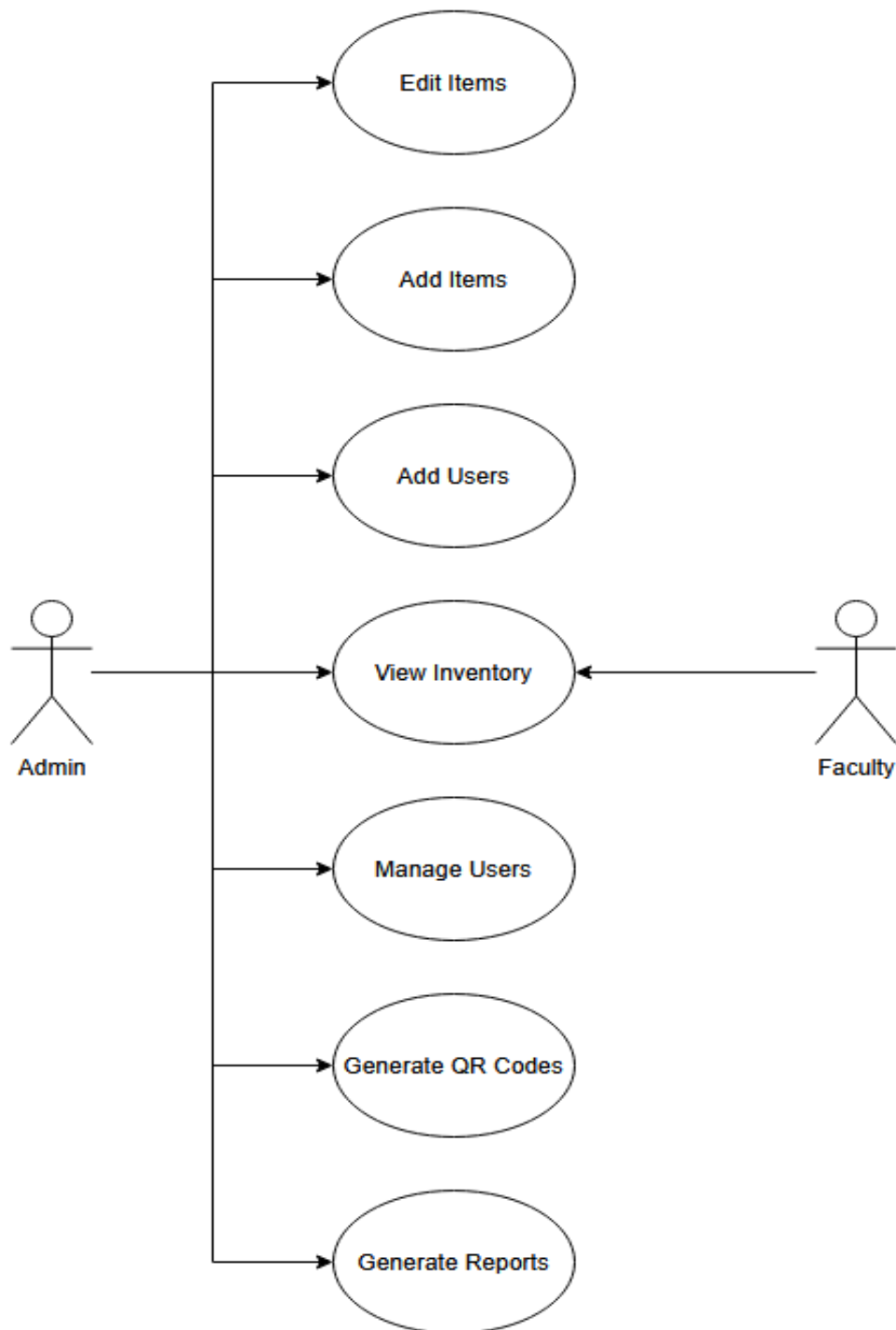


Figure: IPO Diagram for the flow of the Use Case Diagram.

Architectural Framework Diagram

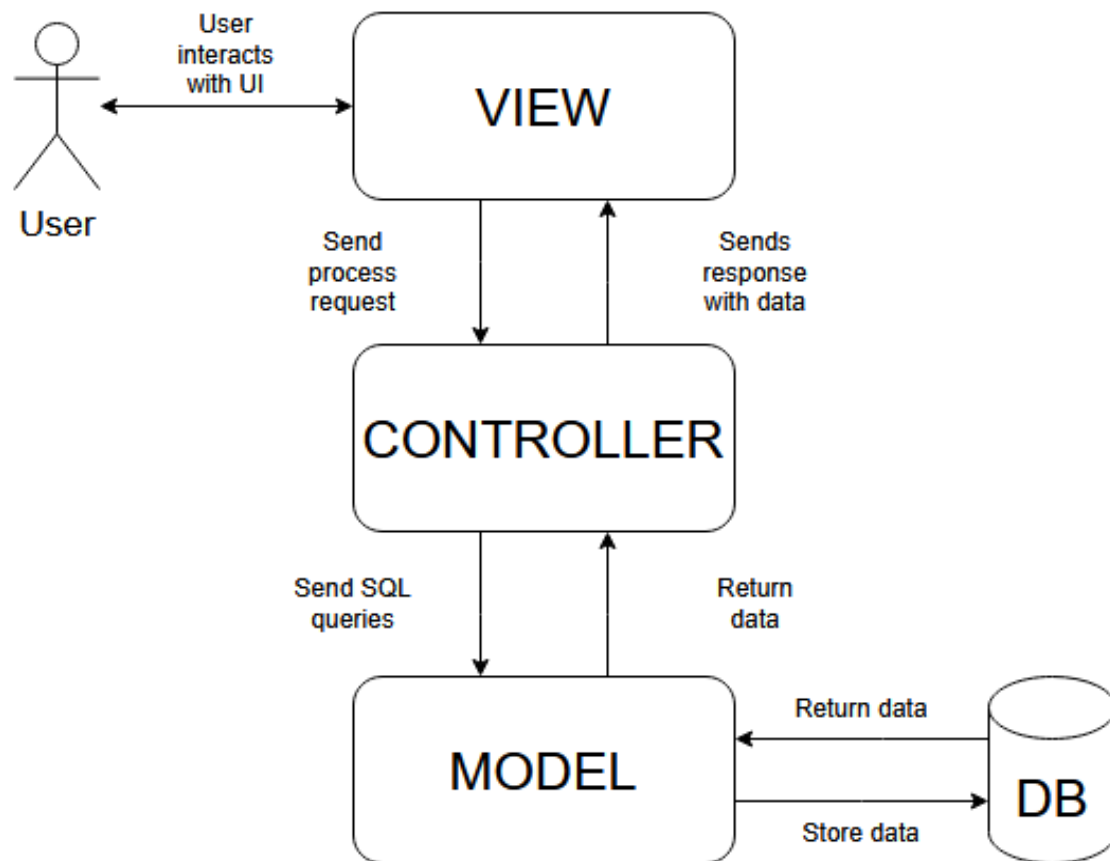


Figure: IPO Diagram for the flow of the Architectural Framework Diagram.

Data Flow Diagram

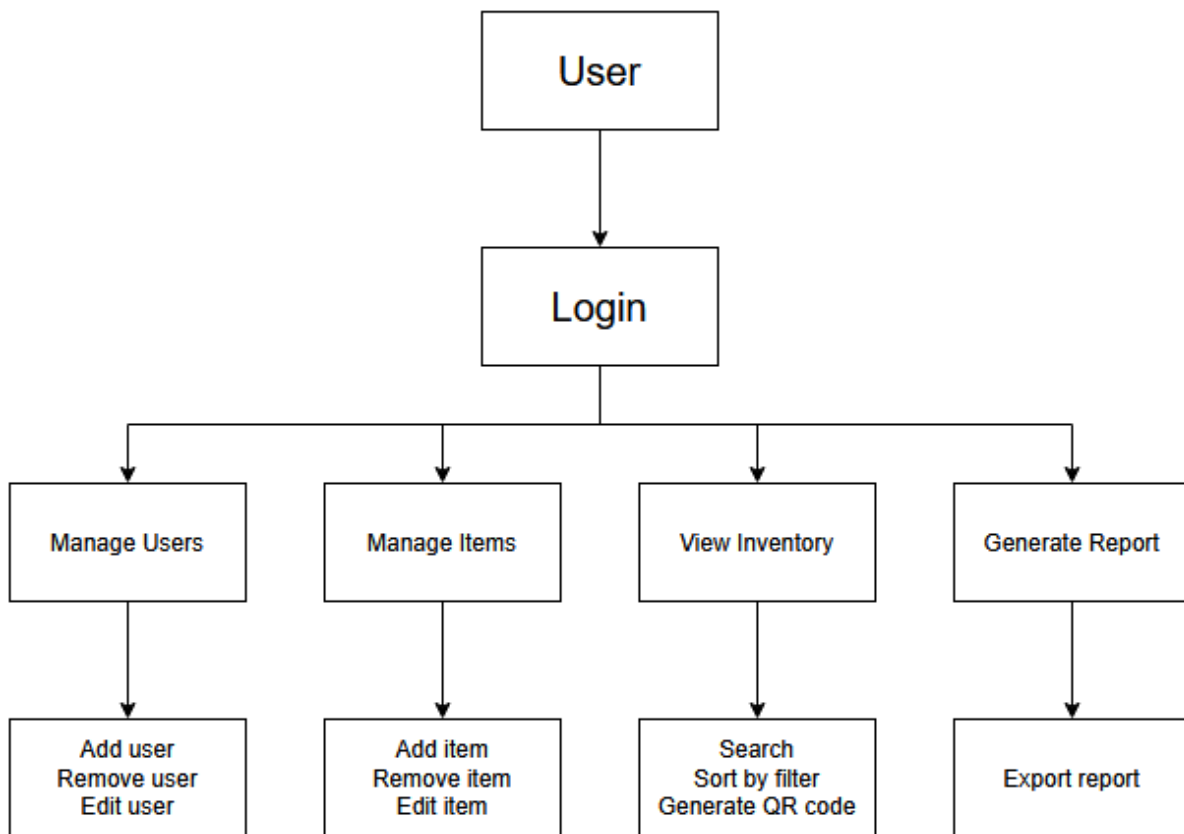


Figure: IPO Diagram for the flow of the Data Flow Diagram.